25901850 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Minor Pathway Courses (MPC) - MG2MPCCHY100 - BASIC ORGANIC CHEMISTRY

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer in Two Sentences

Answer any **10** questions

Each question carries **2** marks

1. Compare the geometry of a carbocation versus a carbanion. [U] / [CO1]
2. Describe the characteristic shape of an sp hybridised carbon atom. [K] / [CO1]
3. Give an example for Anti Markownikoff addition. [K] / [CO2]
4. Identify the electrophile involved in the sulphonation of benzene. [K] / [CO2]
5. Identify the specific reactive intermediate formed during the electrophilic addition of HBr to ethene. [K] / [CO2]
6. Define the term Enantiomer. [K] / [CO3]
7. State Auwers-Skita rule. [K] / [CO3]
8. Draw the most stable and unstable conformation of cyclohexane. [K] / [CO3]
9. Identify the configuration if the two groups of lowest priority are on the same side of the double bond. [K] / [CO3]

10. List two examples of synthetic rubber. [K] / [CO4]

11. Illustrate the simple repeating pattern of a block copolymer using A and B notations. [K] / [CO4]

12. Recall the repeating unit of Polyethylene. [K] / [CO4]

[2x10 = 20]

Part B

Short Answer

Answer any **6** questions

Each question carries **5** marks

13. Write a note on elctrophiles. [U] / [CO1]

14. Write a note on homolytic and heterolytic fission. [U] / [CO1]

15. Compare E1 and E2 reactions. [U] / [CO2]

16. Compare polar and non-polar molecules. [U] / [CO2]

17. Explain why a Meso compound is optically inactive. [U] / [CO3]

18. Identify and Explain the two types of optical rotation- dextrorotatory and laevorotatory. [K] / [CO3]

19. Describe the step-by-step preparation of Nylon 6,6. [U] / [CO4]

20. Classify polymers based on intermolecular forces with suitable examples for each category. [U] / [CO4]

[5x6 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

21. Write an essay on Inductive effect. [U] / [CO1]

22. Write an essay on SN^1 reactions and SN^2 reactions. [U] / [CO2]

23. Describe in detail the conformational analysis of n-butane. [U] / [CO3]

24. Compare and Contrast addition and condensation polymerization. [U] / [CO4]

[10x2 = 20]